React JS

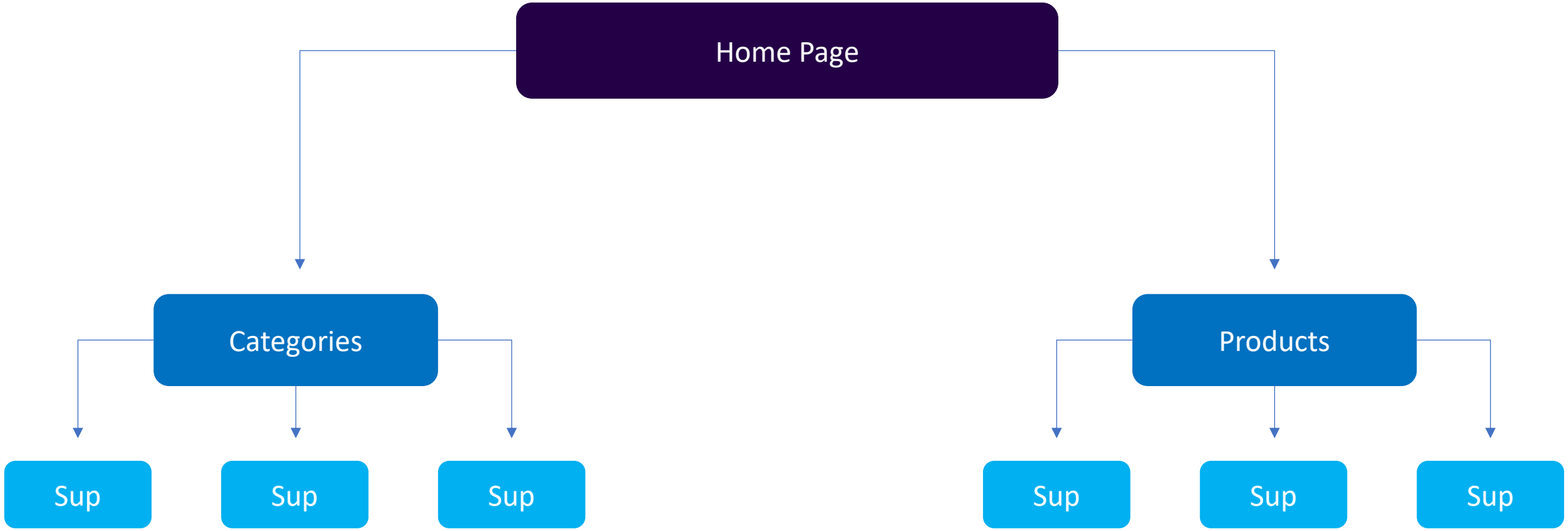
Agenda

- Single Page Application SPA vs MPA
- What is React ?
- Why is React a good choice for developers ?

  JSX | Virtual DOM | Components

- React VS Angular
- What is Node.js & Why we need to use
- Install Requirements & Set up Project
- File Structures
- Integrate with UI libraries

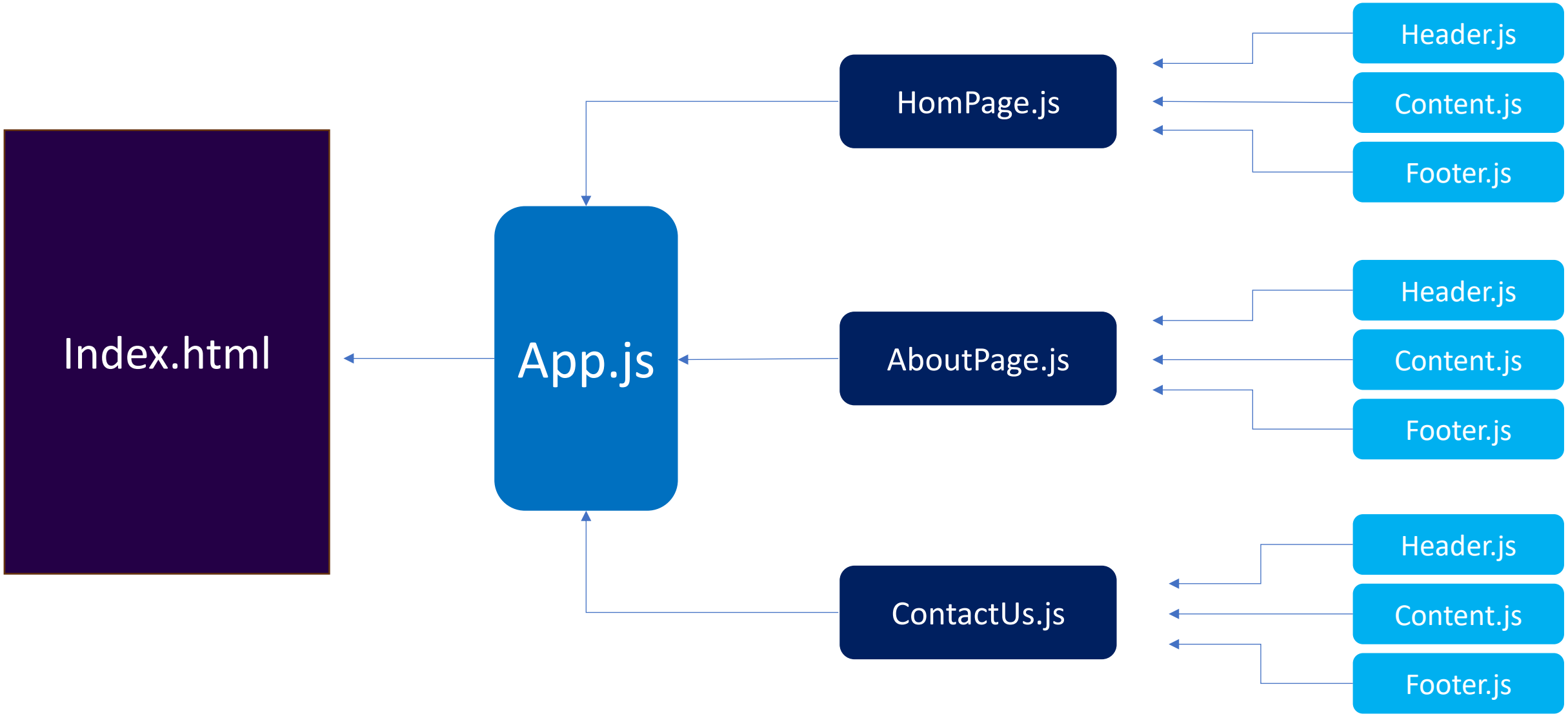
Home
About
Contact
Sign-in
Register
index.
HTML
index.
HTML
index.
HTML
index.
HTML
index.
HTML


Home Page
Categories
Products
Sup
Sup
Sup
Sup
Sup
Sup

✓ Single Page Application (SPA)

- An SPA (Single-page application) is a web app implementation that loads only a single web document, and then updates the body content of that single document via JavaScript code.

- This therefore allows users to use websites without loading whole new pages from the server, which can result in performance gains and a more dynamic experience


Index.html
App.js
HomPage.js
AboutPage.js
ContactUs.js
Header.js
Content.js
Footer.js
Header.js
Content.js
Footer.js
Header.js
Content.js
Footer.js

Single Page Application (SPA)

What is React ??

React is a **java script library** used to create components. it uses **JSX** which is a combination between HTML and Javascript to manipulate DOM elements. React was developed by Facebook in 2011

Why is react a good choice for developers ?? [Benefits]

React is a good choice for developers due to its **reusable components**, fast rendering with the **virtual DOM**, and **strong community** support. It also facilitates the creation of **complex user interfaces** with its declarative approach, making the **code more predictable and easier** to debug.

Components

reusable

Virtual DOM

Community

Complex UI

Clean Code

✓ Why React ?

- Easy creation of dynamic applications.
- Improved performance using Virtual DOM.
- Reusable components
- Easy to learn.
- Dedicated tools for easy debugging.

Component

✓ Component

- React applications are built from isolated pieces of UI called components. A React component is a JavaScript function that you can sprinkle with markup. Components can be as small as a button, or as large as an entire page.
- React component is a JavaScript function that you can sprinkle with markup.
  - ➢ Example : https://miro.medium.com/v2/resize:fit:1400/1*NX0GtVytAI8soIUMAuUSeQ.png

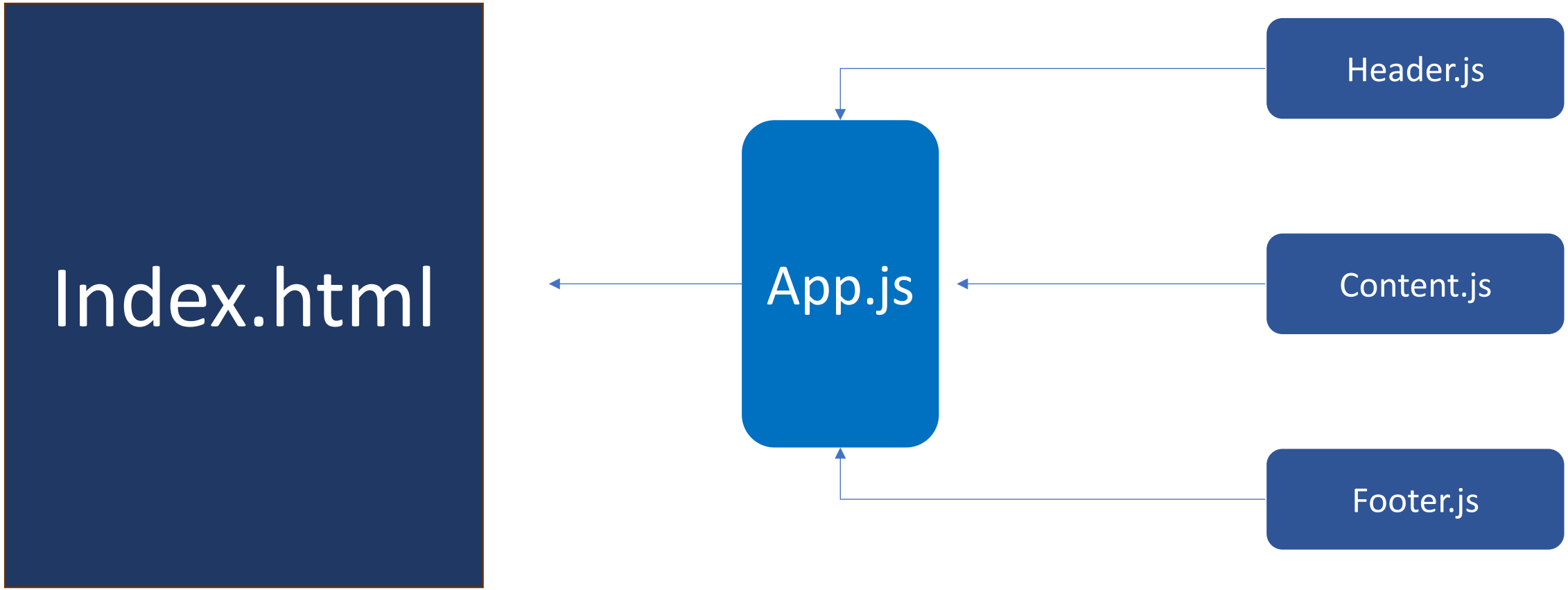
React components are regular JavaScript functions except:

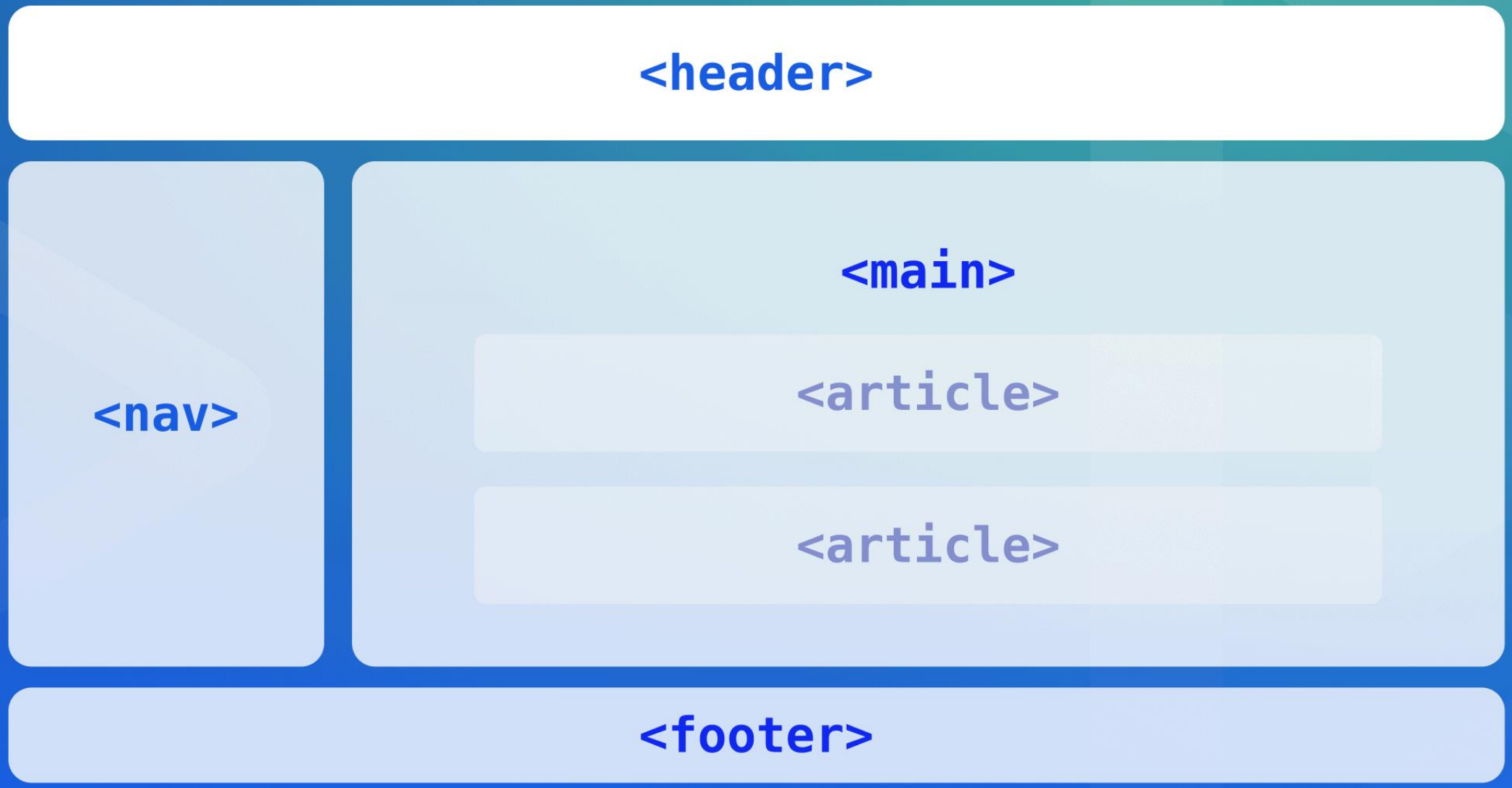
- Their names always begin with a capital letter.
- They return JSX markup.

Header

LOGO

NavBar


Index.html
App.js
Header.js
Content.js
Footer.js

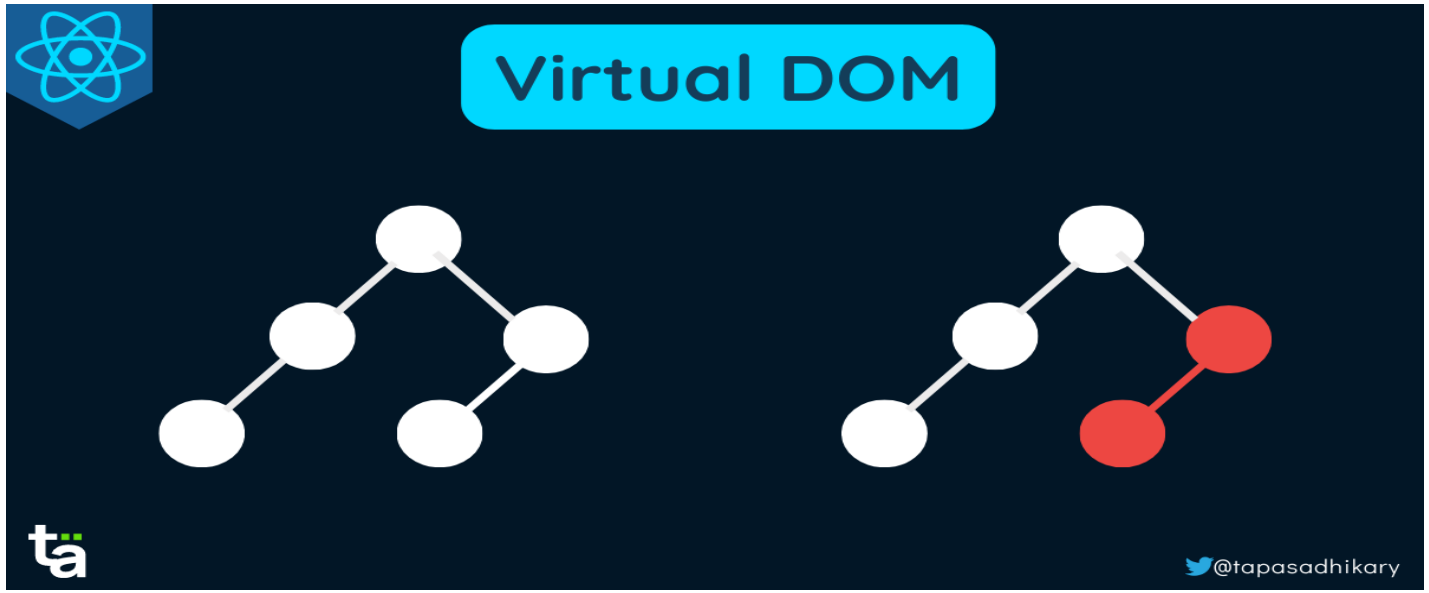


Open Replay

Virtual Dom

Real Dom

Virtual Dom



Add Your Token Code And Get Your Result

25614

Submit

Name	Token Code	Stage	Result
Mustafa Ahmed Mohamed	25614	Third grade	SUCCESSFUL

JSX Code

✓ What is JSX ?

JSX stands for JavaScript XML and allows us to write HTML in React , it is a combination between HTML and Javascript to manipulate DOM elements

✓ JSX Rules

1. **Return a single root element**
   - To return multiple elements from a component, wrap them with a single parent tag.
   - If you don't want to add an extra to your markup, you can write <> and </> instead
2. **Close all the tags**
   - JSX requires tags to be explicitly closed: self-closing tags like <img> must become <img />
3. **camelCase all most of the things!**
   - JSX turns into JavaScript and attributes written in JSX become keys of JavaScript objects.like class would be className

JSX converter: https://transform.tools/html-to-jsx

React VS Angular

library

framework

While both are popular tools for front-end development, React and Angular have significant differences. React is a lightweight **library** focused on the view layer, offering more flexibility and requiring additional libraries for full-fledged application development. On the other hand, Angular is a full-fledged MVC **framework** that comes with a steeper learning curve, but provides a comprehensive set of features out of the box.

Stack overflow Survey :https://survey.stackoverflow.co/2023/#most-popular-technologies-webframe

What is NODE JS?

Node.js is an open-source, cross-platform, back-end JavaScript **runtime environment** that runs on the V8 engine and **executes JavaScript code outside a web browser**. It allows for the development of scalable network applications.

Why do I need Node JS for developing React?

Node JS is essential for developing with React because **it provides the necessary tools** to compile and bundle your React code. Additionally, it enables the use of **npm (Node Package Manager)** for installing and managing the various dependencies required in a React project.

What is React and React-Dom Libs? and what are they used for?

React is a JavaScript library for building user interfaces, particularly single-page applications. It is used for handling the view layer in web and mobile apps. On the other hand, **React-DOM** provides methods for manipulating the Document Object Model (DOM), **allowing** React to **update and render** efficiently in the browser.

What is Babel and why is used in React?

Babel is a JavaScript compiler that **transforms syntax**, allowing you to use next-generation JavaScript today. In React, Babel is used to convert JSX syntax and **ES6**+ code into plain JavaScript that the browser can understand.

✓ Start With Vite

- Install node : https://nodejs.org/en/
- Open a new terminal in your directory.
- Using vite **npm create vite@latest**
- Then select the react app to be installed
- Once app installed, Enter your app directory
- Run the command, **npm install,** then **npm run dev.**

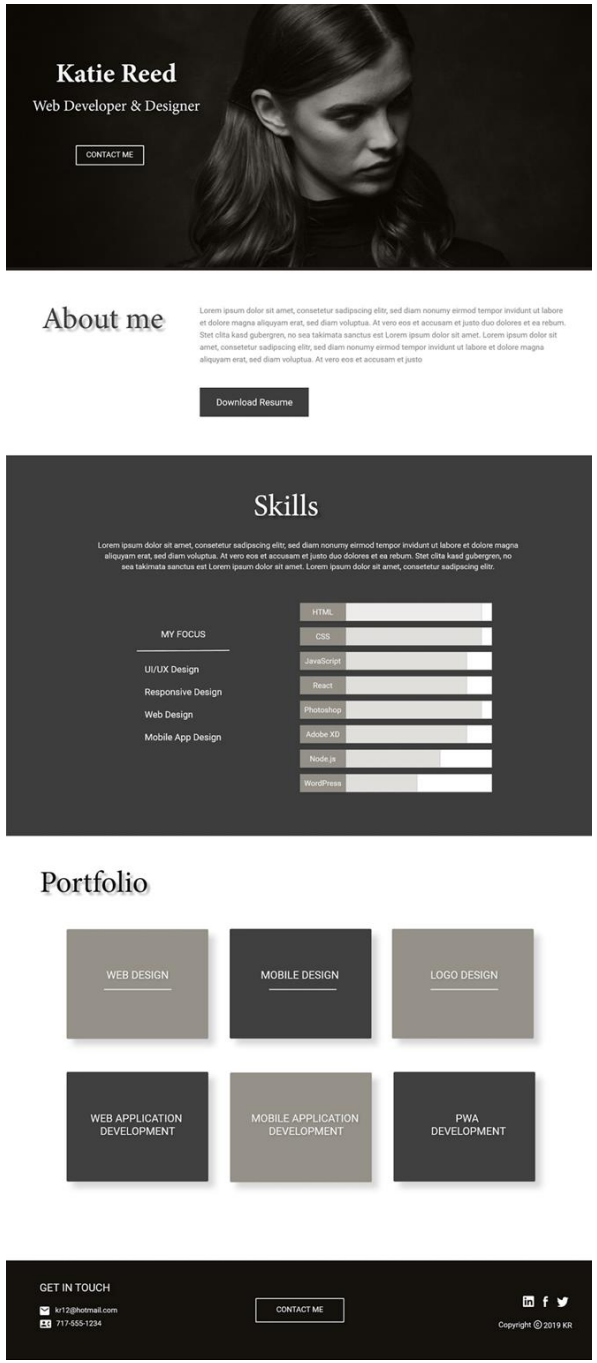
https://vite.dev/guide/

Thank you

lap

Task: Portfolio

Create a web page using React that contains the following sections :

- Hero section
- Bio section , button to download CV.
- Skills section [HTML , CSS …]
- Education Section
- Contacts and social media icon links such as facebook , twitter , linkedin , github
- (fontawesome) - could be in the hero section or footer or the side section as UI.

Using tailwind as UI library (search for it) and create reusable components for your page.